[B19HS3101]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY

Common to CE & IT

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

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S.No.	Question / Problem	CO	KL	M
	UNIT-I			
1.	Define Managerial Economics and Explain its nature and scope	CO1	K2	15
	OR			
2.	What do you mean by Elasticity of demand? Explain in detail about degrees of Price elasticity of Demand?	CO1	K2	15
	UNIT-II			
3.	Define Cost &classify the Elements of Cost?	CO2	K2	15
	OR			
4.	How do you calculate BEP? What are its Assumptions and Applications?	CO2	K3	15
	UNIT-III			
5.	What are Market Structures and explain the features of Perfect Competition?	CO3	K2	15
	OR			
6.	Why is pricing significant in the context of business? Describe any four pricing practices?	CO3	K2	15
	UNIT-IV			
7.	Describe about the Importance of Accounting and types of accounts	CO4	K2	15
	OR			
8.	From the following Trail Balance of Suresh as at December 31, 2013, prepare Trading, Profit and Loss Account for the year ended December 31, 2013 and a Balance Sheet as on that date:	CO4	K3	15

Dr. (Rs.)	Cr. (Rs.)	
Purchases of materials	32,000	
Productive wages	13,000	
Sales	60,000	
Salaries	4,000	
Travelling expenses	1,000	
Carriage inwards	550	
Insurance	300	
Commission	650	
Rent and rates	1,000	
Cash in hand	350	
Cash at bank	5,550	
Repairs	600	
Sundry expenses	110	
Mortgage	6,100	
Buildings	8,000	
Machinery	3,000	
Furniture	1,000	
Stock on hand (1.1.2013)	11,500	
Capital	21,310	
Sundry debtors	9,000	
Sundry creditors	4,200	
	91,610	91,610

Adjust the following:
(a) Prepaid rent Rs. 100
(b) Depreciate the following:
- Buildings @ 10per cent per annum
- Machinery @ 20 per cent per annum
- Furniture @ 15 per cent per annum

(c) Provide for bad debts Rs. 100
(d) Outstanding insurance Rs. 50
(e) Closing stock Rs. 12,000

	UNIT-V			
9.	Explain about capital and the sources available for raising finance	CO5	K2	15
	OR			
10.	Explain about the concept and causes of depreciation. Evaluate the straight-line method and diminishing balance methods.	CO5	K2	15

[B19CE3101]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

STRUCTURAL ANALYSIS II

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

<table>
<tr><td></td><td></td><td></td><td>CO</td><td>KL</td><td>M</td></tr>
<tr><td colspan="6">UNIT – I</td></tr>
<tr><td>1.</td><td></td><td>Using the unit load method, analyse the plain truss shown in the Fig.01 and find the forces in the members.

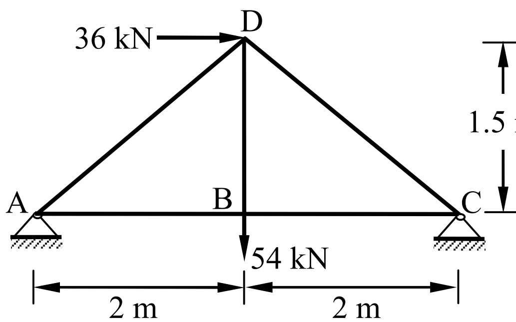



Fig. 01</td><td>1</td><td>K3</td><td>15</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>2.</td><td></td><td>Find the forces in the members of the given truss shown in Fig.02. Cross section area of vertical members is 28 cm^2 and for the other members is 20 cm^2. Take E = 2 × 10^5 MPa. Use castigliano's theorem II.

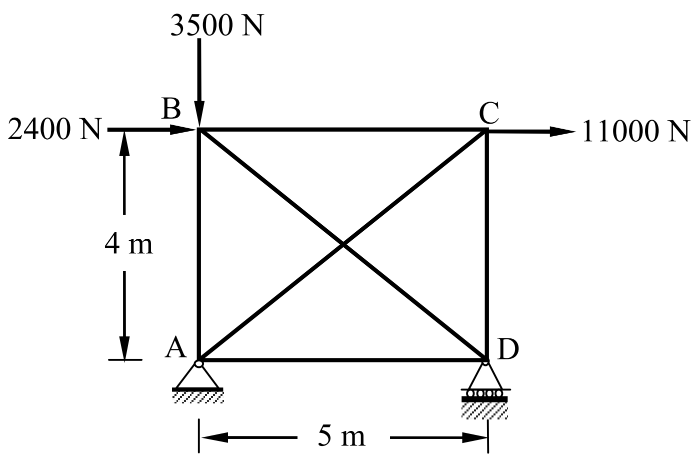
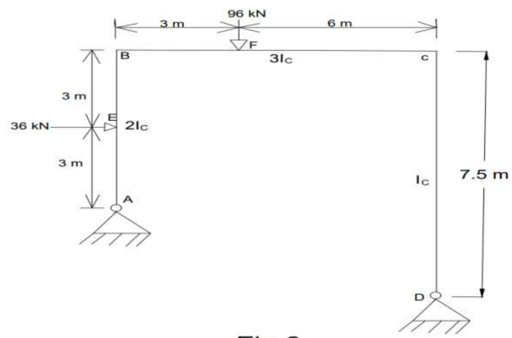
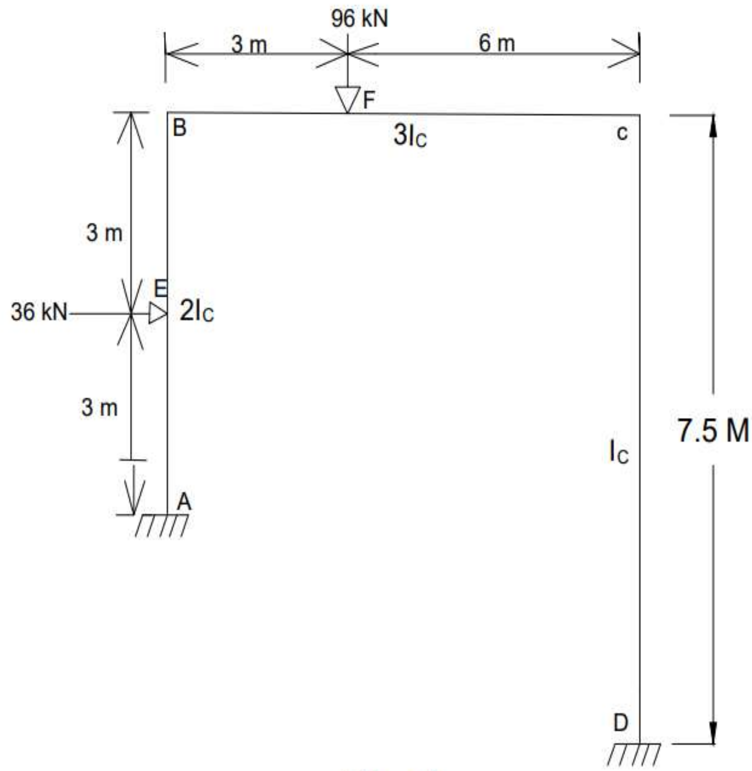
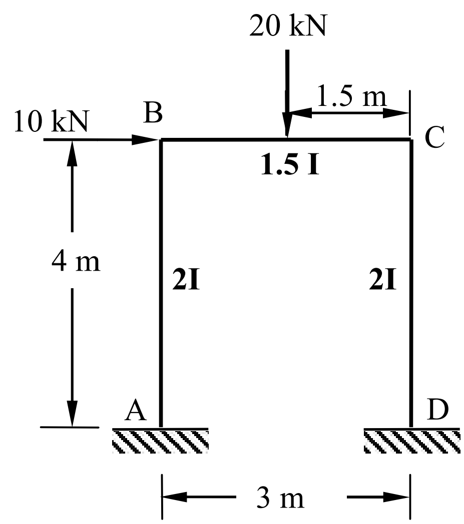
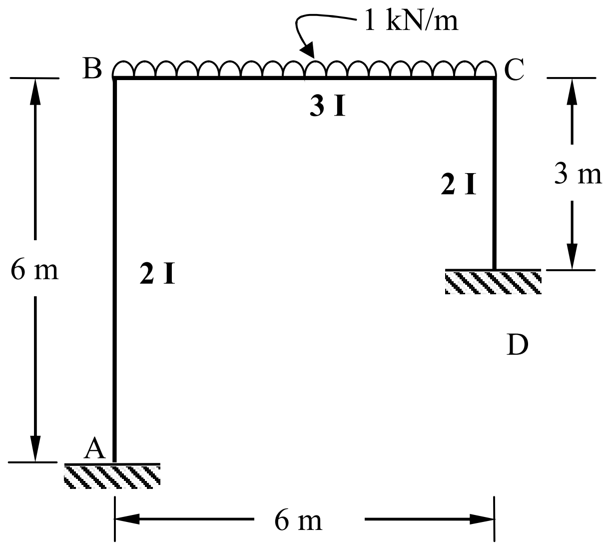



Fig. 02</td><td>1</td><td>K3</td><td>15</td></tr>
<tr><td colspan="6">UNIT – II</td></tr>
</table>

3.		By Method of consistent deformation analyze the rigid frame with two hinged supports shown in Fig.3. Draw free body, shear and moment diagrams and make the equilibrium checks.  Fig.3	2	K3	15
OR					
4.		By theorem of least workanalyze the rigid frame with two fixed supports shown in Fig.3. Draw free body, shear and moment diagrams and make the equilibrium checks.	2	K3	15

		 Fig.4			
		UNIT – III			
5.		Analyse the portal frame shown in Fig.05 by slope deflection method. Draw Bending moment and shear force diagram  Fig.05	3	K3	15
		OR			

6.		Analyse the portal frame shown in Fig.06 by rotation contribution method (Kani's method). Sketch bending moment diagram  Fig. 04	3	K3	15
UNIT – IV					
7.		A symmetrical three hinged parabolic arch of span 40 m and rise 8 m carries a point load of 30 kN at 10 m horizontally from the left hand hinge. The hinges are provided at the supports and at the center of the arch. Calculate the reactions at the supports also calculate the bending moment, radial shear and normal thrust at a distance of 10 m from the left support. Also calculate the maximum Positive and B.M and maximum Negative B.M.	4	K3	15
OR					
8.		A two hinged parabolic arch has span 40 m and rise of 6 m it has second moment of arch varies as secant of the slope of rib axis and carries uniformly distributed load 30 kN/m over left half of the spam together with concentrated load of 120 kN act at 5m from right support. Calculate the reactions and horizontal thrust at the ends and point out the values of maximum positive and negative moments and also find out the radial shear and normal thrust at 10m from right support	4	K3	15
UNIT – V					
9.		A suspension cable hangs between two points A and B, separated horizontally by 100 m. The position of B is 16 m above A. The lowest point in the cable is 3 m below A. The stiffening girder weight 2.5 kN/m and is hinged vertically below the point A,B and the lowest point of the cable . Calculate the maximum tension which occurs in the cable when a 200 kN load crosses the girder from A to B.	5	K3	15
OR					
10.		A suspension bridge of 120 m span has two three hinged stiffening girders supported by two cables having a central dip of 12 m. the road way has a width of 6 m. The dead load on the bridge is 5 kN/m^2 while the live load is	5	K3	15

		10 kN/m^2 which acts on the left half of the span. Determine the shear force and bending moment in the girder at 30 m from the left end. Find also the maximum tension in the cable of position of live load.			

[B19CE3102]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

DESIGN OF REINFORCED CONCRETE STRUCTURES

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a).	Explain characteristic load and design values for materials and loads.	1	K2	5
	b).	Analyze an isolated T-beam, having a span of 6.0 m and cross section having flange width of 1300 mm, flange thickness of 100 mm, web width of 325 mm and an effective depth of 420 mm, is reinforced with 7 Nos of 28 mm dia bars on tension side. The materials used are concrete mix of grade M 20 and HYSDsteelofgradeFe415.	1	K3	10
		OR			
2.	a).	Describe circumstances are doubly reinforced beam used and discuss the advantages of doubly reinforced beams over singly reinforced beams.	1	K2	5
	b).	A rectangular beam of 7m span (center to center between supports) resting on 300mmwidesimplesupports,istocarryauniformlydistributeddeadload(excluding self-weight) of 15 kN/m and a live load of 20 kN/m. Use M20 grade of concrete and Fe 415grade of steel and beam subjected to moderate exposure condition, design the beam section at mid span.	1	K3	10
		UNIT – II			
3.	a).	Explain use of stirrups in a beam.	2	K2	5
	b).	A simply supported beam 300 mm × 600 mm (effective) is reinforced with five bars of 25 mm diameter. It carries a characteristic load of 80 kN/m(including self-weight) over an effective span of 6 m. out of five main bars, two bars can be bent-up safely near the support. Design the shear reinforcement for the beam. Using M 20 grade of concrete and Fe 415 grade of steel.	2	K3	10
		OR			
4.	a).	Discuss the different modes of failure under combined flexure and torsion.	2	K2	5
	b).	Design a rectangular beam section, 300 mm wide and 550 mm deep(overall),subjected to an ultimate twisting moment of 25 kN-m, combined with an ultimate bending moment of 60 kN-m and an ultimate shear force of 50 kN. AssumeM 20 grade of concrete and Fe415 grade of steel.	2	K3	10
		UNIT – III			
5.	a).	Explainthe differencesinthe behavior ofonewayslab and twowayslab.	3	K2	5
	b).	A hall in a building has a floor consisting of a one-way continuous slab cast monolithically with simply supported 250 mm wide beams spaced at 4 m center tocenter. The clear span of the beam is 9 m. Assuming a live load onthe slab as 3 kN/m^2 and floor finish load of 1 kN/m^2. Designthe slab withM20gradeof concreteandFe415grade of steel.	3	K3	10
		OR			
6.	a).	Explain the need for corner reinforcement in two-way rectangular slabs whosecorners areprevented from liftingup.	3	K2	5
	b).	Design a simply supported slab to cover a hall with internal dimensions 4 m ×6 m. The slab is supported on masonry walls 230 mm thick. Assume a live load of	3	K3	10

		3 kN/m^2 and a floor finish load of 1.0 kN/m^2. Use M20 concrete and Fe415 steel. Assume that the slab corners are prevented from lifting up.			
		UNIT – IV			
7.	a).	Distinguishbetween(i)unsupportedlengthandeffectivelengthofacompressionmember; (ii) bracedcolumn and un-bracedcolumn.	4	K2	5
	b).	Design the reinforcement for a column with l_{ex}= l_{ey}= 3.5 m and size 300 mm ×500 mm, subject to a factored axial load of 1250 kN with biaxial moments of180 kN-m, and 100 kN-m (i.eM_{ux}= 180 kN-m, M_{uy}= 100 kN-m). Assume M 25grade of concreteandFe415 grade of steel.	4	K3	10
		OR			
8.	a).	Expressthe functionsofthetransversereinforcementinreinforced concrete columns.	4	K2	5
	b).	Design a short square column, with effective length 3.0 m capable of safety resisting the following factored load effects (under uniaxial eccentricity) = 1625 kN, M_u = 75 kN-m Assume M25 grade of concrete and Fe 415 grade of steel.	4	K3	10
		UNIT – V			
9.	a).	Explainthe typesofshear checkstobedoneinisolated footings.	5	K2	5
	b).	Design an isolated footing for a rectangular column 300 mm × 500 mm, reinforcedwith 6-25 mm ø bars,and carrying afactoredaxial load of 1000 kNandafactored uniaxial moment M_{ux} = 120 kN-m(with respective to the majoraxis) at the column base. Assume the moment is reversible.Assume soil withanallowable pressure of 200kN/m^{2}ata depthof1.25mbelow ground.Assume Fe415 grade of steel for column and footing and M20 grade of concrete forthefootingand M25gradeof concrete for column.	5	K3	10
		OR			
10.	a).	Distinguish structural behaviour between 'stair slabsspanningtransversely'andstair slabsspanninglongitudinally.	5	K2	5
	b).	Design a dog-legged staircase('waist slab') for an office building, assume a floor tofloor height of 3.0 m, a flight width of 1.2 m, and a landing width of1.25 m. Assume the stairs to be supported on 230 mm thick masonry wall at the edges of the landing, parallel to the risers. Assume live loads of 5.0 kN/m^{2}and mild exposure conditions. Use M20 grade of concrete and Fe415 grade of steel.	5	K3	10

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[B19CE3103]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II B. Tech II Semester (R19) Regular Examinations

SOIL MECHANICS

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a	Derivethe Relation $\rho_d = \frac{(1-n_a)G.\rho_w}{1+WG}$	1	K3	7
	b.	In a field exploration, a soil sample was collected in a sampling tube of internaldiameter5cm belowgroundwatertable.Thelength of theextracted samplewas 10.2cmand its weight was 387gm.If G=2.7,and the weight of the driedsample is 313gm,Determine the porosity, Void Ratio, Degree of Saturation,Bulk densityand drydensityofthe sample.	1	K3	8
		OR			
2.	a.	DiscussaboutIndianStandardClassificationSystem	1	K3	7
	b.	A soil sample obtained from a core cutter is 500mm in diameter and 100mm in height. Its wet unit weight is 17kN/m3 and dry unit weight is 13kN/m3. If the specific gravity of solids is 2.8 determine the volume of solids of the specimen and also determine the moisture content in the soil specimen.	1	K3	8
		UNIT – II			
3.	**a.**	ASanddepositconsistsoftwolayers.The toplayeris2.5mtthick(ρ=1709.67Kg/m3)andthebottomlayeris3.5mthick (ρ_{sat}=2064.52Kg/m3).The water table is at a depth of 3.5m from the surface and the Zone of capillary saturation is 1 m above the water table. Determine the Total, Neutral and effective stresses.	2	K3	8
	b.	Explainthe factors EffectingPermeability	2	K3	7
		OR			
4.	a.	ExplainQuicksandconditionandcriticalhydraulicgradient.	2	K3	7
	b.	A Constant head Permeability test was carried out on a cylindrical sample of sand 10cms dia& 15cm height 160cm^3 of water was collected in 1.75 mins. Under a head of 30cm. Compute the Coefficient of permeability (K) & Velocity of Flow.	2	K3	8
		UNIT – III			
5.	a.	Derive an expression for the vertical stress at a point due to a point load, using Boussinesq's theory.	3	K3	7

	b.	A square foundation5mx 5m is to carrya load of 4000kN.Calculate thevertical stress at a depth of 5m below the centre of the foundation IN=0.084for m=n=0.5. Also determine the vertical stress using 2:1 distribution.	3	K3	8
OR					
6.	a.	Explainabout Newmark's InfluenceChart	3	K3	7
	b.	AConcentrated Loadof 2000kNisappliedat the groundsurface.Determinetheverticalstress atapoint P which is6m directlybelow theload. Also calculatetheverticalstressatapointRwhichisatadepthof6mbutatahorizontaldistance of 5m from theaxis of the load.	3	K3	8
UNIT – IV					
7.	a.	What is the Effect of Compaction on Engineering Properties of Soil? Discuss inbriefly	4	K3	7
	b.	A Stratum of Clay is 2m thick and has an initial overburden pressure of50kN/m2 at its middle. Determine the final settlement due to an increase of 40kN/m2 at the middle of the clay layer. The clay is over-consolidated, with apre consolidation pressure of 75kN/m2.The values of the coefficients of recompression and Compression index are 0.05 and 0.25, respectively. Take initial Void ratio as 1.40	4	K3	8
OR					
8.	a.	DiscussTerzaghi'stheoryof consolidation,Statingvariousassumptions	4	K3	7
	b.	A3m thick clay layer beneatha building isover lain by apermeable stratum and is under lain by an impervious rock. Thecoefficient of consolidation of the clay was found to be 0.025cm2/minute.Thefinal expected settlement for the layeris 8cm,(a) How much time will ittake for 80% of the total settlement to take place? (b)Determine the time required for a settlement of 2.5cm to occur (c)Compute the settlement that would occur in one year.	4	K3	8
UNIT – V					
9.	a.	Whataretheadvantagesoftriaxialshear testover thedirect sheartest	5	K3	7
	b.	In a direct shear test, the following results are obtained. Normal stress (kN/m2)25 50 75 Shear stress at failure (kN/m2) 30 45 60 Size of soil sample is 60mm x 60mm. Find the shear parameters if a triaxial test was conducted on the same soil with a cell pressure of 30kN/m2, what would be the deviator stress at failure?	5	K3	8
OR					
10.	a.	How the shear tests classified is based on drainage conditions? What practical situations do they simulate?	5	K3	7
	b.	A shear Vane of 7.5cm diameter and 11.00cm length was used to measure the shear strength of soft clay. If a torque of 600N-m was required to shear the soil, Calculate the shear strength. The Vane was then rotated rapidly to cause remolding of the soil. The Torque Required in the remoulded state was 200N- m.Determine the Sensitivity of the soil.	5	K3	8

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[B19CE3104]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

ESTIMATION, SPECIFICATIONS & CONTRACTS

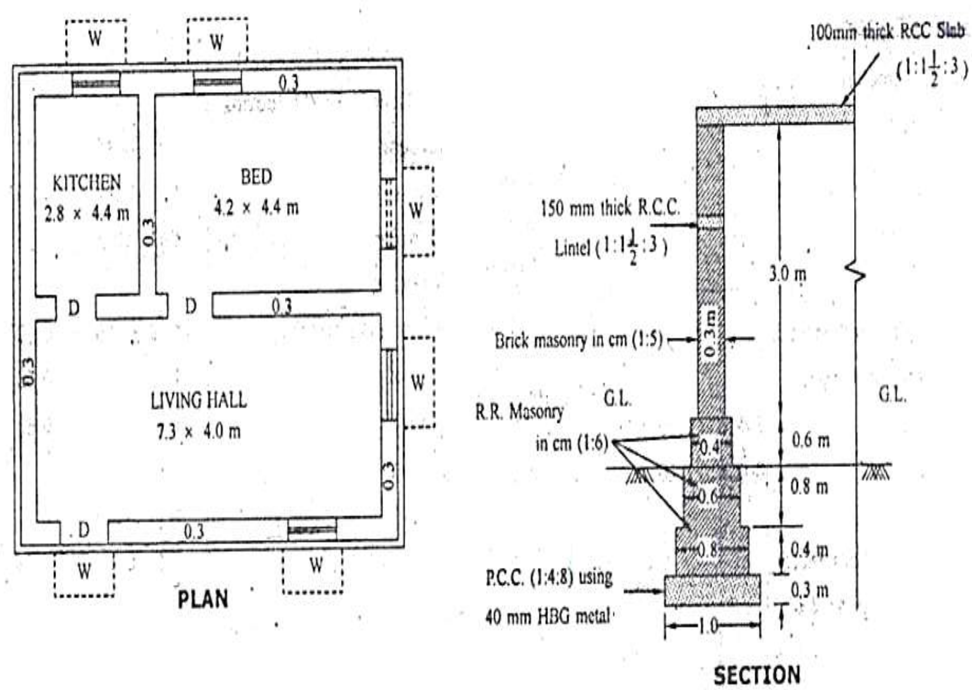
MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

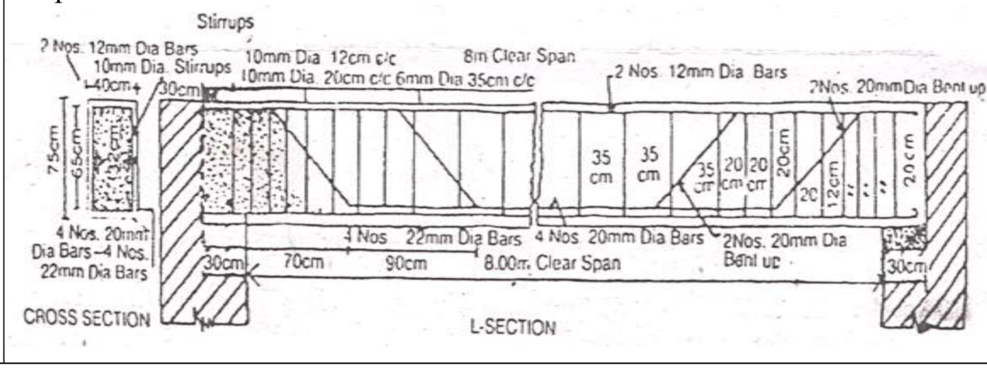
All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a	State and explain different types of Sanctions/Approvals	CO1	K2	7
	b	Explain the following: i) SoR ii) Contingencies and Work charged establishment	CO1	K2	8
		OR			
2.	a	Explain about data required for preparation of estimate?	CO1	K2	7
	b	State and explain different types of estimates?	CO1	K2	8
		UNIT – II			
3.	a	Estimate the quantities of the building items of a two roomed building from the given plan and section as shown in the figure below i) Lime concrete in foundation ii) 1st class brick work with 1:6 cement mortar in foundation iii) Brick work in Super structure	CO2	K3	15



		OR			
4.	b	Prepare a detailed estimate of the following items for a R.C.C. beam of 8 meter clear span and 75cm X 40cm in section from the given figure	CO2	K3	15

	b	R.C.C. work including centering and shuttering Steel in detail shall be calculated separately and Prepare a schedule of bars.			



UNIT – III

5.	a	Prepare the rate analysis for Brick masonry work in CM (1:6).	CO3	K3	7
	b	Prepare the rate analysis forPlastering in CM (1:4), 12mm thick for walls	CO3	K3	8

OR

6.	a	Give the detailed specifications for the following items of work i) RCC (1:2:4) for roof slabs ii) Random Rubble stone masonry in (1:3) lime sand mortar	CO3	K2	8
	b	Give the detailed specifications for the following items of work i) Plastering for walls ii) Wood work for door and window frames	CO3	K2	7

UNIT – IV

7.	a	Estimate the cost of earthwork for a portion of road for 360m length from the following data: The formation width of the road is 10m and the side slopes are 2:1 in banking and 1.5:1 in cutting	CO4	K3	15

Stn	25	26	27	28	29	30	31	32	33	34	35
Dist (m)	0	40	80	120	160	200	240	280	320	360	400
Rlof Gl (m)	51.0	50.9	50.5	50.8	50.6	50.70	51.2	51.4	51.3	51.0	50.6
Rlof Fl (m)	52.0	---------------Downward gradient of 1 in 200----------------------→									

OR

8.	b	The ground levels along the center line of the road are given below	CO4	K3	15

Chainage (meters)	0	50	100	150
RL of Ground	97.00	96.50	96.00	97.50

The road is to be formed in embankment with the formation level at

		100.00m throughout the length. If the road width is10.00 m and the side slopes 2:1, calculate the quantity of earthwork required by Trapezoidal rule. Assume transverse slope as level			
UNIT – V					
9.	a	Differentiate between i) Market Value and Book Value ii) Scrap value and Salvage Value	CO5	K2	8
	b	An old building has been purchased by a person @ a cost of Rs. 6,00,000/- excluding the cost of land. Calculate the amount of annual sinking fund @ 9% interest assuming the life of building as 30 years and the scrap value of the building as 10% of the purchase.	CO5	K3	7
OR					
10.	a	Define contact and explain types of contracts.	CO5	K2	8
	b	Explain about the conditions of contract in construction	CO5	K3	7

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[B19CE3105]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

REPAIR, REHABILITATION AND RETROFITTING OF STRUCTURES

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a	Define the following terms i) Repair, ii) Retrofit, iii) Rehabilitation, iv) strengthening	CO1	K3	8
	b	What are the causes of deterioration of concrete? Explain any three.	CO1	K3	7
		OR			
2.		Explain the causes and characteristics of cracks in RCC structural members.	CO1	K3	15
		UNIT – II			
3.	a	Discuss the purpose of damage assessment.	CO2	K3	7
	b	List out various Non-destructive testing methods and explain any two of them.	CO2	K3	8
		OR			
4.	a	Explain about the rapid assessment of damage.	CO2	K3	7
	b	Explain the chemical tests of concrete.	CO2	K3	8
		UNIT – III			
5.	a	Explain the effects due to temperature on durability of concrete	CO3	K3	8
	b	Explain the mechanism of corrosion of steel reinforcement.	CO3	K3	7
		OR			
6.		Mention the various corrosion protection techniques and explain any three.	CO3	K3	15
		UNIT – IV			
7.	a	Explain brieflyCFRP and GFRP	CO4	K3	8
	b	Explain about sulphur infiltrated concrete	CO4	K3	7
		OR			
8.	a	Explain the usage of epoxy resins.	CO4	K3	8
	b	Explain about ferro cement as rehabilitation material.	CO4	K3	7
		UNIT – V			
9.	a	Explain about drypack repairing technique	CO5	K3	7
	b	Differentiate between shoring and underpinning. Explain any two methods of underpinning	CO5	K3	8
		OR			
10.	a	Explain the column jacketing technique with neat sketch	CO5	K3	7
	b	List out various techniques for strengthening of beams and explain any one technique with neat sketch.	CO5	K3	8

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[B19CE3106]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

II B. Tech I Semester (R19) Regular Examinations

GEOSYNTHETICS AND ITS APPLICATIONS

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a.	Explain the characteristics of different types of geosynthetics.	1	K2	8
	b.	Explain about geosynthetic clay liners and geocells.	1	K2	7
		OR			
2.	a.	Explain in detail about geotextiles and geomembranes.	1	K2	8
	b.	Explain in detail any two major factors to be consider in the selection of a geosynthetic for field application.	1	K3	7
		UNIT – II			
3.		Describe the major steps of manufacturing process for Woven and Non-woven geotextiles	2	K2	15
		OR			
4.		Explain various bonding processes used in manufacturing of geosynthetics	2	K2	15
		UNIT – III			
5.		Explain in detail the mechanical properties of geosynthetics.	3	K2	15
		OR			
6.		Explain various hydraulic properties of geosynthetics in detail.	3	K2	15
		UNIT – IV			
7.		Explain the reinforcement and filtration function of geosynthetics in detail.	4	K2	15
		OR			
8.	a.	Explain drainage function of geosynthetics in detail.	4	K2	8
	b.	Explain the suitability of various geosynthetics according to their function.	4	K2	7
		UNIT – V			
9.		Explain in detail any two applications of geosynthetics in soil reinforcement.	5	K2	15
		OR			
10.		Explain how geosynthetics are used for hydraulic applications.	5	K2	15

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[B19CE3107]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

ADVANCED WATER AND WASTE WATER ENGINEERING

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	(a)	Explain reverse osmosis process with a neat sketch.	CO1	K2	8
	(b)	Explain how colour and odour are removed from water.	CO1	K2	7
		OR			
2.	(a)	Explain the working of a grit chamber with a neat sketch.	CO1	K2	8
	(b)	Explain the nano-filtration technique used in water treatment.	CO1	K2	7
		UNIT – II			
3.	(a)	Explain what is meant by backwashing of filters.	CO2	K2	8
	(b)	Compare plain sedimentation with sedimentation with coagulation.	CO2	K2	7
		OR			
4.	(a)	Explain granular media filtration in detail.	CO2	K2	8
	(b)	Explain types of chlorination.	CO2	K2	7
		UNIT – III			
5.	(a)	Explain aerobic and anaerobic processes of treatment.	CO3	K2	8
	(b)	Explain the suspended growth process used for wastewater treatment.	CO3	K2	7
		OR			
6.	(a)	Explain the working principle of hybrid growth systems	CO3	K2	8
	(b)	Compare batch reactors with continuous type reactors.	CO3	K2	7
		UNIT – IV			
7.	(a)	Explain the working and operation of MBBR with a neat sketch.	CO4	K2	8
	(b)	Explain the working principle of an stabilization pond.	CO4	K2	7
		OR			
8.	(a)	Explain the working and operation of Sequential Batch Reactor (SBR).	CO4	K2	8
	(b)	Explain how hybrid reactors help in stabilization of waste water.	CO4	K2	7
		UNIT – V			
9.	(a)	Explain the working and operation of UASB reactor with a neat sketch.	CO5	K2	8
	(b)	Explain how sludge is handled from its generation point to diposal,	CO5	K2	7
		OR			
10.	(a)	Explain the working of an anaerobic filter with a neat sketch.	CO5	K2	8
	(b)	Explain how sludge is digested and disposed.	CO5	K2	7

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[B19CE3108]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

OPEN CHANNEL FLOW

MODEL QUESTIONPAPER

TIME: 3Hrs **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a.	Explain the classification of flow in channels.	1	K2	8
	b.	Find the velocity of flow and rate of flow of water through a rectangular channel of 6m wide and 3m deep, when it is running full.The channel is having bed slope as 1 in 2000. Take chezys constant C=55.	1	K3	7
		OR			
2.	a.	Derive the discharge equation through open channel by Chezys formula.	1	K3	8
	b.	Find the discharge through a rectangular channel 14m wide, having depth of water 3m and bed slope 1 in 1500. Take N= 0.03 in Kutters formula.	1	K3	7
		UNIT – II			
3.	a.	Derive the equation for most economical rectangular channel having width b and depth d.	2	K3	8
	b.	A rectangular channel carries water at the rate of 400 litres/sec when bed slope is 1 in 2000. Find the most economical dimensions of the channel if C=50.	2	K3	7
		OR			
4.	a.	Derive the most economical circular section for maximum velocity condition.	2	K3	8
	b.	Determine the maximum discharge of water through a circular channel of diameter 1.5m when the bed slope of channel is 1 in 1000. Take C=60.	2	K3	7
		UNIT – III			
5.	a.	Derive the equation of minimum specific energy in terms of critical depth.	3	K3	8
	b.	Find the specific energy of flowing water through a rectangular channel of width 5m when the discharge is $10m^3$/sec and depth of water is 3m.	3	K3	7
		OR			
6.	a.	Explain Specific energy curve with a neat figure.	3	K3	8
	b.	The specific energy for a 5m wide rectangular channel is to be 4Nm/N. If the rate of flow through the channel is $20m^3$/sec, determine the alternate depths of flow.	3	K3	7
		UNIT – IV			
7.	a.	Derive the expression for depth of Hydraulic Jump.	4	K3	8
	b.	The depth of flow of water, at a certain section of a rectangular channel of 4m wide, is 0.5m. The discharge through the channel is $16m^3$/sec. If a hydraulic jump takes place on the downstream side, find the depth of flow after the jump.	4	K3	7
		OR			

8.	a.	Explain about surges in open channels.	4	K2	8
	b.	Explain the types and applications of Hydraulic Jump.	4	K2	7
UNIT – V					
9.	a.	Derive the equation of Gradually Varied Flow.	5	K3	8
	b.	Find the slope of the free water surface in a rectangular channel of width 20m, having depth of flow is 5m. The discharge through the channel is $50m^3$/sec. The bed of the channel is having a slope of 1 in 4000. Take chezys constant C=60.	5	K3	7
OR					
10.	a.	Derive the expression for the length of the back water curve.	5	K3	8
	b.	Determine the length of the back water curve caused by an afflux of 2m in a rectangular channel of width 40m and depth 2.5m. The slope of the bed is 1 in 11000.Take Mannings N= 0.03.	5	K3	7

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[B19CE3201]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

DESIGN OF STEEL STRUCTURES

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

Use of IS: 800:2007 and Steel tables are allowed

All questions carry equal marks

			CO	KL	M
UNIT – I					
1.	a	Explain (i) Pitch of bolt (ii) Gauge distance (iii) Edge distance	1	K2	3
	b	Design a lap joint between two plates of size 120 ×12 mm thick and 100 × 12 mm thick using a single row of M20 bolt of grade 406 and grade 410 plates.	1	K4	12
OR					
2.	a	Give any two factors that cause failure of bolted joints.	1	K2	3
	b	Two plates of 200 × 12 mm are to be connected by a double cover bolt joint with 20 mm diameter bolt. The factored tensile force on the plates is 500 kN. Design the bolted connection.	1	K4	12
UNIT – II					
3.	a	Explain weld defects.	2	K2	3
	b	Design a connection to joint two plates of size 200 × 10 mm of grade Fe 410 to use full plate tensile strength using shop fillet welds if (i) a lap joint is used (ii) a double cover bolt joint is used.	2	K4	12
OR					
4.	a	Explain inspection methods of welds.	2	K2	3
	b	An ISMC 250 is used to transmit a factored force of 700 kN. The channel section is connected to a gusset plate 10 mm thick. Design a fillet weld, if the overlap is limited to 300 mm. Use slot welds if required.	2	K4	12
UNIT – III					
5.		Design a tension member 3.4 m between c/c of intersections using double angle section and carrying a factored pull of 200 kN. The member is subjected to reversal of stresses.	3	K4	15
OR					
6.		Design a tension member to carry a factored tensile load of 400 kN. Two angles placed back-to-back with long legs outstanding are desirable. The length of the member is 2.9 m.	3	K4	15
UNIT – IV					
7.		Design a laced column 9 m long to carry a factored axial load of 1200 kN. The column is fixed at both the ends. Provide single lacing system with bolted connection. The column consist of two channels placed back-to-back.	4	K4	15
OR					

8.		Design a built-up column 9 m long to carry a factored axial compressive load of 1100 kN.The column is restrained in position but not in direction at both the ends. Design the column with connecting system as battens with bolted connections. Use two channel sections back-to-back. Use steel of grade Fe 410.	4	K4	15
UNIT – V					
9.		A Simply Supported steel joist of 5.0 m span has to support a load of 60 kN/m (inclusive of self weight). The beam compression flange is restrained against buckling. Design an appropriate section using steel of grade Fe 410.	5	K4	15
OR					
10.		Design a Simply supported beam of span 3.5 m subjected to a factored bending moment of 300 kNm and factored shear of 140 kN. The beam is laterally unsupported. steel grade of Fe 410.	5	K4	15

[B19CE3202]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

TRANSPORTATION ENGINEERING – I

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

		UNIT – I	**CO**	**KL**	**M**
1.	(a)	Briefly outline the highway development in India.	1	K2	**7**
	(b)	Compare Nagpur & Bombay Road Development plans.	1	K2	**8**
		OR			
2.	(a)	What is master plan? Explain the importance of master plan in highway planning.	1	K2	**7**
	(b)	Briefly outline about rural road development plan?	1	K2	**8**
		UNIT – II			
3.	(a)	Discuss the design factors of Highway alignment?	2	K2	**7**
	(b)	Calculate the safe stopping distance for design speed of 50 kmph for two-way traffic on a two lane road. Assume coefficient of friction as 0.35 and reaction time of driver as 2.5 seconds.	2	K4	**8**
		OR			
4.	(a)	Derive an expression for extra widening on horizontal curvers.	2	K4	**7**
	(b)	Find out the length of transition curve for the following data: Radius of horizontal curve = 400m, design speed = 100 kmph, length of wheel base = 6.2m, number of lanes = 2, rainfall at the location = heavy, terrain condition = hilly, superelevation is introduced by rotating the edges with reference to center line and rate of introduction of superelevation is 1 in 150. Width of highway is 7m.	2	K4	**8**
		UNIT – III			
5.	(a)	Explain how the speed and delay studies are carried out. What are the various uses of delay studies?	3	K2	**7**
	(b)	What are the various types of parking facilities designed for traffic needs?	3	K2	**8**
		OR			
6.	(a)	Explain the relationship between speed, travel, time, volume, density and capacity.	3	K2	**7**
	(b)	Explain the various measures that may be taken to prevent accidents.	3	K2	**8**
		UNIT – IV			
7.	(a)	Explain the test procedure of shape test for aggregates.	4	K2	**7**
	(b)	Estimate the thickness of cement concrete pavement using the method suggested by IRC. Modulus of elasticity of concrete = 3.5 X 105 kg/cm2, Modulus of rupture of concrete = 30 kg/cm2 , Poisson's ratio of concrete = 0.17, Modulus of subgrade reaction = 5 kg/cm2 , Wheel load = 4800 kg, Radius of contact area = 13 cm.	4	K3	**8**
		OR			
8.	(a)	Discuss the properties of bitumen.	4	K2	**7**
	(b)	Explain group index method of pavement design. What are the limitations of the method?	4	K2	**8**
		UNIT – V			
9.	(a)	Write a descriptive note on maintenance of highways.	5	K2	**7**
	(b)	Write a short note on alligator and reflection cracking.	5	K2	**8**

OR					
10.	(a)	What are the requirements of material, plants & equipment’s for bituminous pavement construction? Discuss briefly.	5	K2	**7**
	(b)	Give a descriptive note on Pavement Evaluation Techniques.	5	K2	**8**

[B19CE3203]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

FOUNDATION ENGINEERING

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a.	Explain two boring methods with neat sketch	1	K2	7
	b.	Describe,in brief,various geo physical methods.	1	K2	8
		OR			
2.	a.	Discuss standard penetration test. What are the various corrections?	1	K2	7
	b.	ThemeasuredN valuein a boreholeisfound tobe20.The depth of theborehole was15m.Thewatertable was located1.5mbelowgroundlevel. Thesoilisfine siltyanditsunitweightaboveandbelow watertablewas16and21kN/m3respectively.Calculate the corrected value of N.	1	K3	8
		UNIT – II			
3.	a.	Whataredifferenttypesofshallow foundation? Explainwiththehelp of sketches.	2	K2	7
	b.	ASquarefootingistobedesignedtocarryatotalloadof500KNinasoil having$C=10kN/m^2$,$Ø=25^0$,$\gamma=16kN/m^3$.Thefootingistobelaidatadepthof 2mts.belowgroundlevelandF.Sis3.Find theDimensionofthefooting.Given $N_c=22$,$N_q=11$,$N_\gamma=9$.	2	K3	8
		OR			
4.	a.	Derive the expression for Terzaghi's Bearing Capacity theory and write the assumptions?	2	K3	7
	b.	A footing 4mx2minplan,transmitsapressureof150N/sq.m on a cohesive soil having E=6x104 kN/sq.m and$\mu=0.50$.Determine the immediate settlement of the footing at the center, assuming it to be (a) a flexible footing, I=1.32 and (b) a rigid footing,I=1.20	2	K3	8
		UNIT – III			
5.	a.	Explain Pneumatic Caisson with neat sketch and mention its advantages and disadvantages.	3	K2	7
	b.	A concrete pile, 25cm in diameter is driven in to a dense sand (Ø = 300 , γ=20kN/m3 ,K=1, Tanδ=0.70) for a depth of 6m. If the water table is at 2m from the ground level, estimate the safe load, taking Factor of safety 3.	3	K3	8
		OR			
6.	a.	Discuss the Causes and Remedies for Tilts and Shifts (with neat sketches) in Well Foundations.	3	K2	7
	b.	A pile group consists of 9 piles of 300mm dia and length 10m, is arranged in 3 rows at spacing of 750mm c/c. The piles are driven into a deposit of clay whose properties are C=100kN/m2 , γ=20kN/m3 .Determine the safe load on the pile group. Take F.S=3,α=0.6,Nc=9	3	K3	8

UNIT – IV					
7.	a.	Discuss the Friction circle method for the stability analysis of slopes. Can this method be used for purely cohesive soil?	4	K2	7
	b.	A cut of length 10m is made in a cohesive soil deposit (C=30kN/m2 , Ø = 00 , γ=20kN/m3).There is a hard stratum under the cohesive soil at a depth of 12m below the original ground surface. If the required factor of safety is 1.5, determine the safe slope	4	K3	8
OR					
8.	a.	Explain types of slope failures	4	K2	7
	b.	A slope is to be Construction in a soil For which C=0and Ø = 360. It is to be assumed that the water level may occasionally reach the surface of a slope, with seepage taking place parallel to the slope. Determine the maximum slope angle for a factor ofsafety1.5,assuming a potential failure surface parallel to the slope What would be the factor of safety of the slope, constructed at this angle ,if the water tables should be well below the surface?	4	K3	8
UNIT– V					
9.	a.	What are the assumptions of Rankine's theory? Derive the expressions for active pressure and passive pressure	5	K3	7
	b.	Determine the passive pressure by Rankine's theory per unit run for a retaining wall 4m high, with i=150 , Ø = 30 0 and γ=20kN/m3 . The back face of the wall is smooth and vertical.	5	K3	8
OR					
10.	a.	Discuss Culmann's method for the determination of active earth pressure.	5	K2	7
	b.	A5mhigh rigid retaining wall has to retain a backfill of dry cohesionless soil having the following properties: Ø = 30^0, e=0.74 and G=2.68. Plot the distribution of Rankine lateral earth pressure on the wall and determine the magnitude and point of application of the resultant thrust.	5	K3	8

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[B19CE3204]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

AIR POLLUTION AND CONTROL

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	(a)	Define air pollution. Explain the various factors influencing air pollution.	CO1	K2	8
	(b)	Explain about the typical gaseous pollutants and their sources	CO1	K2	7
		OR			
2.	(a)	Explain various air quality standards	CO1	K2	8
	(b)	Explain the major harmful effects of particulate matter on human health and materials?	CO1	K2	7
		UNIT – II			
3.	(a)	Explain the plume behaviour in detail.	CO2	K2	8
	(b)	What is maximum mixing depth? Explain its practical significance in Air pollution control.	CO2	K2	7
		OR			
4.	(a)	What is atmospheric stability? Explain the temperature inversions with reference to air pollution.	CO2	K2	8
	(b)	What is meant by effective stack height? Mention guidelines for minimum.	CO2	K2	7
		UNIT – III			
5.	(a)	List the recorded major air pollution episodes chronologically. State their significant features.	CO3	K2	8
	(b)	How SOx and NOx emissions affect human health and the growth of plants?	CO3	K2	7
		OR			
6.	(a)	Explain the effects of carbon monoxide and sulphur oxides on human health.	CO3	K2	8
	(b)	Explain any two air pollution disasters occurred during 20^{th} and 21^{st} centuries.	CO3	K2	7
		UNIT – IV			
7.	(a)	List the sampling devices used for air quality monitoring. Explain in brief various sampling methods of gaseous pollutants.	CO4	K2	8
	(b)	Explain the importance of stack monitoring. List out various methods of stack monitoring.	CO4	K2	7
		OR			
8.	(a)	Write short notes on Ambient Air Quality monitoring.	CO4	K2	8
	(b)	What is the use of high volume sampler? Explain.	CO4	K2	7
		UNIT – V			
9.	(a)	Explain with the help of a neat sketch the working of an electrostatic precipitator.	CO5	K2	8

	(b)	Explain with the help of a neat sketch the working of a centrifugal scrubber.	CO5	K2	7
OR					
10.	(a)	Explain working of a cyclone separator with a neat sketch.	CO5	K2	8
	(b)	Explain when do you recommend absorption as a method of control of gaseous contaminants?	CO5	K2	7

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[B19CE3205]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

PRESTRESSED CONCRETE

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a	Explain about pre-tensioning and post-tensioning systems with figures.	1	K2	8
	b	Mention advantages and limitations of prestressed concrete?	1	K2	7
		OR			
2.	a	Explain briefly principal of post-tensioning?	1	K2	8
	b	Explain the types of prestressing systems and explain Freyssinet system.	1	K2	7
		UNIT – II			
3.	a	A pre tensioned beam of rectangular cross-section, 150 mm wide and 300 deep, is pre stressed by 8.7 mm wires located 100 mm from the soffit of the beam. If the wires are initially tensioned to a stress of 1100 N/mm^2, calculate their stress at transfer and the effective stress after all losses given the following data. Up to time of transfer Total Relaxation of steel --- 35 N/mm^2 70 N/mm^2 Shrinkage of concrete --- 100×10^{-6} 300×10^{-6} Creep Coefficient --- --------- 1.6 E_s = 210 kN/mm^2 E_c = 31.5 kN/mm^2	2	K3	10
	b	Explain types of losses in prestressed concrete beams?	2	K2	5
		OR			
4.	a	A pretensioned beam 250 mm wide and 300 mm deep is prestressed by 12 wires each of 7 mm diameter initially stressed to 1200 N/mm^2 with their centroids located 100 mm from the soffit, estimate the final percentage loss of stress due to elastic deformation, creep, shrinkage and relaxation using IS:1343-1980 code and the following data: Relaxation of steel stress = 90N/mm^2, Es = 210N/mm^2, Ec = 35 N/mm^2, Creep coefficient ø =1.6, Residual shrinkage strain = 3×10^{-4}	2	K3	10
	b	Explain creep of concrete in prestressed members?	2	K2	5
		UNIT – III			
5.	a	A rectangular concrete beam,100 mm wide by 250 mm deep, spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m. a) Calculate the resultant stress distribution for the central cross-	3	K3	15

		section of the beam. The density of concrete is 24 kN/m^3. b) Find the magnitude of the prestressing force with an eccentricity of 40 mm which can balance the stresses due to D.L and L.L at the bottom fibre of the central section of the beam.			
OR					
6.		A prestressed concrete beam of section 120 mm wide by 300 mm deep is used over an effective span of 6 m to support a uniformly distributed load of 4 kN/m, which includes the self-weight of the beam. The beam is prestressed by a straight cable carrying a force of 180 kN and located at an eccentricity of 50 mm. Determine the location of the thrust line in the beam and plot its position at quarter and central span sections?	3	K3	15
UNIT – IV					
7.	a	A pretensioned, T-section has a flange which is 300 mm wide, 200 mm thick. The rib is 150 mm wide by 350 mm deep. The effective depth of the cross-section is 500 mm. Given, Ap= 200 mm^2, f_{ck}= 50 N/mm^2 and f_u=1600 N/mm^2, estimate the ultimate moment capacity of theT-section using the Indian Standard Code regulations.	4	K3	10
	b	Explain about the assumptions made in analysis of flexure?	4	K2	5
OR					
8.	a	A prestressed concrete beam of rectangular section 300 mm wide and 600 mm deep is prestressed by two post tensioned cables of area 600 mm^2each initially stressed to 1600 N/mm^2. The cables are located at a constant eccentricity of 100 mm. The span of the beam is 10 m. If f_{ck}= 40 N/mm^2. Estimate the ultimate shear resistance of support section uncracked in flexure.	4	K3	10
	b	Explain shear recommendations of IS 1343-1980	4	K2	5
UNIT – V					
9.		The end block of a post–tensioned prestressed member is 550 mm wide and 550 deep. Four cables, each made up of 7 wires of 12 mm diameter strands and carrying a force of 1000 kN, are anchored by plate anchorages, 150 mm×150 mm, located with their centres at 125 mm from the edges of the end block. The cable duct is of 50 mm diameter. The 28 days cube strength of concrete f_{cu} is 45 N/mm^2. The cube strength of concrete at transfer f_{ci} is 25 N/mm^2. Permissible bearing stresses behind anchorages should confirm with IS 1343. The characteristics yield stress in mild steel anchorage reinforcement is 260 N/mm^2. Design suitable anchorage for the end block.	5	K3	15
OR					

10.	a	Describe the stress distribution in end block in post-tensioned member.	5	K2	5
	b	A Freyssinet anchorage (125 mm diameter),carrying 12 wires of 7 mm diameter stressed to 950N/mm^2,is embedded concentrically in the web of an I-section beam at the ends.The thickness of the web is 225 mm. Evaluate the maximum tensile stress and the bursting tensile force in the end block.Design the reinforcement for the end block?	5	K3	10

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[B19CE3206]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

GROUND IMPROVEMENT TECHNIQUES

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT – I					
1.	a.	Explainvariousfactorsthataffectfieldcompactionofsoils?	1	K2	7
	b.	Explainindetailthedifferenttypesofdrainsusedforin-situdensificationof cohesivesoilswithneatsketches?	1	K2	8
OR					
2.	a.	Describevariousmethodsofinstallationofstonecolumns?	1	K2	7
	b.	Explainvarioustypes of rollers usedinfieldcompaction ofsoils.	1	K2	8
UNIT – II					
3.	a.	Explaindifferenttypesofgroutsgivingone exampletoeach.	2	K2	7
	b.	Explainvariouscategoriesofgroutingwithneatsketcheswherevernecessary.	2	K3	8
OR					
4.	a.	Explainvariouscomponents ofgroutplant.	2	K2	7
	b.	Whatis Tube-A-Manchette.Howgroutingis done usingit.	2	K2	8
UNIT – III					
5.	a.	Explainvariousapplicationsofgeotextilesinaccordance withtheirfunctions.	3	K2	7
	b.	Explainanytwotestscarriedoutforsuitabilityofgeotextilesasreinforcement insoil.	3	K2	8
OR					
6.	a.	Explaindifferenttestsyouwouldconducttodeterminethephysicaland hydraulicpropertiesofgeotextiles?	3	K2	15
UNIT – IV					
7.	a.	Whatisreinforcedsoil.Howisitdifferentfromreinforcedconcrete?	4	K2	7
	b.	Explainindetail thecomponentsof Reinforcedsoil	4	K2	8
OR					
8.	a.	Whatarethevariousapplicationsofreinforced soil?	4	K2	7
	b.	Explainvariousfactorsthataffectangleofinterfacialfriction.	4	K2	8
UNIT – V					
9.	a.	Explain Ruthfutch’s Method of proportioning soils for Mechanical stabilization?	5	K2	7
	b.	What islime stabilization of soils?Explain variousengineeringbenefits of lime stabilizationofsoils.	5	K2	8
OR					
10.	a.	Whatiscementstabilizationofsoils?Explainsoil-cementreactionsandfactors thateffectcement stabilizationofsoil.	5	K2	7
	b.	Explainvariousin-situ methodsusedin stabilizationofsoils.	5	K2	8

[B19CE3207]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B.Tech II Semester (R19) Regular Examinations

ENVIRONMENTAL IMPACT ANALYSIS

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	(a)	List the elements of EIA and explain in detail.	CO1	K2	7
	(b)	Name the guidelines for preparation of EIS and explain.	CO1	K2	8
		OR			
2.	(a)	List governmental policies for protection of environment and discuss.	CO1	K2	8
	(b)	Define EIS. Explain it in detail.	CO1	K2	7
		UNIT – II			
3.	(a)	Discuss different environmental indices in detail.	CO2	K2	8
	(b)	Explain the socio-economic attributes and their effect on environment.	CO2	K2	7
		OR			
4.	(a)	Explain the human and cultural aspects.	CO2	K2	7
	(b)	Explain different indices and their importance in EIA study.	CO2	K2	8
		UNIT – III			
5.	(a)	Explain different methodologies used for EIA study.	CO3	K2	8
	(b)	Explain the criteria for selection of methodology in EIA assessment.	CO3	K2	7
		OR			
6.	(a)	Explain matrix and adhoc methodologies in detail.	CO3	K2	7
	(b)	Explain checklist and network methods in detail.	CO3	K2	8
		UNIT – IV			
7.	(a)	Explain how do you predict the impact on air and water	CO4	K2	7
	(b)	Write about prediction and assessment of human and aesthetic attributes in EIA	CO4	K2	8
		OR			
8.	(a)	Write in detail how do you predict the impacts of socio economic aspects	CO4	K2	8
	(b)	Explain the prediction and assessment of noise and air	CO4	K2	7
		UNIT – V			
9.	(a)	Explain the significance of cost-benefit analysis.	CO5	K2	8
	(b)	Explain the impact of any thermal power plant on environment.	CO5	K2	7
		OR			
10.	(a)	Explain about controlling measures of environmental impacts.	CO5	K2	8
	(b)	Explain the impact of mining industry on environment.	CO5	K2	7

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[B19CE3208]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

HYDRAULIC STRUCTURES

MODEL QUESTIONPAPER

TIME: 3Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a.	Explain the step by step procedure of determining the reservoir capacity by mass inflow curve.	1	k2	8
	b.	Explain the engineering investigations required for reservoir planning.	1	k2	7
		OR			
2.	a.	Explain various methods of apportionment of total cost of a multi-purpose reservoir.	1	k2	8
	b.	Explain the Zones of storage in reservoir.	1	k2	7
		UNIT – II			
3.		Explain the forces acting on a gravity dam with a neat sketch.	2	K2	15
		OR			
4.	a.	Discuss in detail foundation treatment of a gravity dam.	2	K2	8
	b.	Design the practical profile of a gravity dam of stone masonry for the following data: RL of base of dam= 1450m, RL of H.F.L=1480.5m Specific gravity of the masonry= 2.4, Height of waves= 1m Safe compressive stress for masonry= 120t/m^2	2	K3	7
		UNIT – III			
5.	a.	Explain the criteria for safe design of earth dam.	2	K2	8
	b.	With the help of sketches explain in detail how seepage is controlled in earthen dams.	2	K2	7
		OR			
6.	a.	Explain the causes of structural failures of earthen dams.	2	K2	8
	b.	Explain the stability analysis of earth dams.	2	K2	7
		UNIT – IV			
7.	a.	Define Stilling Basin. Explain USBR and IS Stilling Basins.	3	K2	8
	b.	Discuss various methods used for energy dissipation below spillways.	3	K2	7
		OR			
8.	a.	Explain the types of spillways.	3	K2	8
	b.	Explain the components of spillways.	3	K2	7
		UNIT – V			
9.	a.	Explain the method of independent variables for estimating the pressures under impervious floors of weirs.	4	K2	8
	b.	Differentiate between Blighs creep theory and Khoslas theory.	4	K2	7
		OR			
10.		Explain the design of Vertical drop weir with a neat figure.	4	K2	15